

# STARTING AT THE BEGINNING: New Perspectives on the Biology of Mucosal T Cells

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■ **Abstract** The gastrointestinal tract is the central organ for uptake of fluids and nutrients, and at the same time it forms the main protective barrier between the sterile environment of the body and the outside world. In mammals, the intestine has further evolved to harbor a vast load of commensal bacteria that have important functions for the host. Discrimination by the host defense system of nonself from self can prevent invasion of pathogens, but equivalent responses to dietary or colonizing bacteria can lead to devastating consequences for the organism. This dilemma imposed by the gut environment has probably contributed significantly to the evolutionary drive that has led to sophisticated mechanisms and diversification of the immune system to allow for protection while maintaining the integrity of the mucosal barrier. The immense expansion and specialization of the immune system is particularly mirrored in the phylogeny, ontogeny, organization, and regulation of the adaptive intraepithelial lymphocytes, or IEL, which are key players in the unique intestinal defense mechanisms that have evolved in mammals.

## INTRODUCTION

The intestine of higher vertebrates is a major T cell organ. Three defined anatomical compartments contain T lymphocytes. The gut-associated lymphoid tissues (GALT), including Peyer's patches (PP) and mesenteric lymph nodes (MLN), harbor T cells in organized lymphoid structures. T cells residing in the GALT have the most in common with peripheral lymph node T cells. In the other two compartments, T lymphocytes are scattered among nonlymphoid cell types. These include T cells distributed within the lamina propria (lamina propria lymphocytes, or LPL). Additionally, numerous intraepithelial lymphocytes (IEL) reside on the other side of the basement membrane from the lamina propria. IEL are found as single cells between the intestinal epithelial cells, with about one lymphocyte for every four to nine epithelial cells in the small intestine. It has been estimated, based on immunohistology, that the IEL of the mouse small intestine alone amount to almost 50% of the total T cell number in all lymphoid organs (1), although the number

that can be purified is somewhat less. Additionally, the T cells residing in these different intestinal compartments are significantly different from conventional T cells as well as from one another in terms of ontogeny, phenotype, and function, and it is therefore evident that the IEL provide qualitatively and quantitatively major contributions to the biology of the mucosal and systemic immune systems.

This review discusses the biology of IEL, with an emphasis on the subsets that express an  $\alpha\beta$  TCR. Although they are heterogeneous, as a group IEL exhibit a number of properties that distinguish them from peripheral T cells. The distinct nature and complexity of IEL is undoubtedly dictated by their unique location at the critical interface between the intestinal epithelial barrier and the gut lumen.

## THE IMMUNE FUNCTION ORIGINATED IN THE GUT

Host defense by specialized immune mechanisms arose hundreds of millions of years ago from the nondefensive, food-absorbing mechanism of phagocytosis by single-cell organisms. With the evolution of complex, multicellular organisms, differentiation and specialization became apparent, and food-absorbing cells became distinct from other cell types. The initial “gastral layer” of phagocytic cells first appeared in *Metazoans* and showed much homology with the gastrointestinal epithelial cell layer of the higher vertebrates. When the function of phagocytosis assumed its defensive role, protection of the organism from self-destruction initiated the first immune recognition. Defensive phagocytosis most likely only occurred in the absence of self-recognition, established via inhibitory antiself receptors. This self-recognition-based defense system had much in common with inhibitory NK receptors, which are regulated by recognition of self-MHC class I molecules. In fact, NK-like receptors appeared before T and B cell antigen receptors, as evidenced by the finding of a homolog of the C94/NKR-P1 NK-type receptor in the urochordate *Botryllus* (2).

The origin of the adaptive immune system probably goes back to the gastrointestinal regions of the first “jawed” fish in the *Placoderms* species some 500 mya. The intestine as a central primary immune organ is evident in primitive vertebrates including the sea lamprey and hagfish (3, 4), but this is less apparent in higher vertebrates. The dominant evolutionary drive that led to the expansion and specialization of immune defense mechanisms coincided with the increasing complexity of the organism itself and the constant race to keep up with the evolving and increasing number of pathogens. This could have been the evolutionary selection for the acquisition of the ancestral *RAG* gene for antigen-receptor gene recombination, which is likely to have arisen from an ancient bacterial transposon (5). A particularly strong selective pressure might have been imposed on the multiple tasks of the digestive tract with the dietary changes concurring with the emergence of the jaw function and the migration of these early vertebrates from water to land. This might have led to the first need for compartmentalization of the intestinal tract (6). The mechanisms controlling this differentiation and specialization of

the gastrointestinal tract are poorly understood, but homeobox-containing genes, such as *Hox-1.4* (7) and *Hox-3* (8), have been implicated in this differentiation process.

## THE EVOLUTION OF THE THYMUS

Although the gastrointestinal tract has retained some of its ancestral ability for lymphopoiesis, particularly for B-lymphocytes, the thymus has evolved as the primary lymphoid organ for T cell development. The appearance of the thymus seems to coincide with the sudden appearance of a family of MHC antigen-presenting molecules and expanding repertoires of adaptive T and B cell antigen receptors. This dramatically increased the capacity of the immune system to cope with the staggering demand for recognition of nonself external molecules, and with the complexity of the organism itself, as well as the need to control for neoplasm. Paradoxically, this sudden expansion also increased the potential of self-destruction by the immune system, and strategies to educate immune cells became a necessity (9).

The evolutionary origin of the thymus is unknown, but several phylogenetic as well as ontogenetic facts suggest a connection to the gastrointestinal tract. In humans, the primordial thymus develops from the anterior portion of the embryonic gut tube (the pharyngeal pouches). At one point during gestation, the third pair of pouches buds off the gut epithelium (endodermis) and forms a sac-like epithelial protrusion that bends toward the digestive tract. This becomes the thymus rudiment. As the thymus compartment grows, it pushes into the mesoderm that surrounds the primordial gut, and some of the mesenchymal tissue becomes incorporated into the thymus tissue. Subsequently, it becomes colonized by bone marrow-derived lymphocyte precursors, macrophages, and DCs, all mesodermal in origin. The thymus epithelium further develops and differentiates from the gut endoderm, but only with the stimulus from mesenchymal cells. The lymphocyte precursors interact with the epithelium in a two-way fashion, whereby the epithelial cells induce lymphoid differentiation, and at the same time they themselves differentiate further by the stimulus of developing lymphoid cells (10, 11). As a result, the thymus tissue becomes organized and a clear cortex and medulla are formed. Unlike the typical layer of gut epithelium cells with tight connections and limited intercellular spaces, the thymic epithelial cells are loosely connected. Nevertheless, desmosomes or tight junctions remain and are witnesses of their epithelial origin (12, 13). It has been suggested recently that thymus epithelium is composed of a variety of cells with different antigen-presenting capacity and which display a mosaic of ectopically expressed self-antigens (14). Speculations have been put forward that this developmental plasticity of the thymus organ may be important for the generation of self-tolerance to some tissue-specific antigens, and we propose here that this ectopic expression of organ-specific self-antigens is equally important for the selection in the thymus of self-specific T cells that are plentiful among the IEL (discussed below).

Thymocyte precursors are believed to arrive in the subcapsular region of the thymic cortex, where the initial events of conventional positive selection occur. The selected thymocytes subsequently move to the medulla for further maturation and differentiation, and eventually they exit the thymus. The thymus organ involutes later in life, when a peripheral repertoire of mature  $\text{TCR}\alpha\beta^+$  T cells and memory T cells is established and the need for newly generated naïve T cells is no longer desirable. As a consequence in older animals, the remnant lymphopoietic capacity of the gut occasionally becomes more apparent (15).

## MODERN TIMES: MUCOSAL T CELL SUBSETS IN MAMMALS

Mammalian IEL are almost exclusively T cells, but they have diverse phenotypes with regard to TCR and coreceptor expression, as well as the expression of other cell surface molecules. Unlike peripheral T cells, however, all the diverse populations of IEL display an antigen-experienced phenotype. Nevertheless, the locations and pathways the various IEL subsets follow to encounter antigens, and the nature of these antigens, distinguish them from one another. There are three broad categories of IEL: the subset of  $\gamma\delta\text{TCR}^+$  IEL and two distinct subsets of  $\alpha\beta\text{TCR}^+$  IEL, those that express one of the conventional TCR coreceptors, CD4 or CD8 $\alpha\beta$ , and those that lack coreceptor expression and are double negative (DN). A common feature of all IEL subsets is the unique capacity to express CD8 $\alpha\alpha$ , a characteristic of the activated nature of the mucosal T cells and their adaptation to the gut environment.

The ratio of  $\gamma\delta$  to  $\alpha\beta$  TCRs in IEL varies among species. **In mice, a significant portion of the IEL in the small intestine expresses a  $\gamma\delta\text{TCR}$ , and they predominate in young animals at three weeks of age, after which both  $\text{TCR}\gamma\delta^+$  and  $\text{TCR}\alpha\beta^+$  IEL increase in numbers (16).** They are less prevalent in the large intestine (17, 18), and in humans  $\text{TCR}\gamma\delta^+$  IEL are a relatively small ( $\sim 10\%$ ) population (19), but their numbers increase significantly during coeliac disease (20).  $\gamma\delta$  TCRs generally do not recognize processed peptides presented by MHC class I or class II molecules (21, 22), and  $\gamma\delta$  TCRs share properties with both adaptive receptors encoded by rearranging genes and invariant NK-like receptors. Similar to the self-based NK receptors, some  $\gamma\delta\text{TCRs}$  can interact directly with self-MHC molecules such as the mouse T10 and T22 molecules (23) and human MIC molecules (24), which are not loaded with processed antigens. The specificity of most mouse  $\gamma\delta\text{TCR}$  IEL, however, is unknown.  $\gamma\delta$  T cells are also capable of sensing stressed cells by a direct recognition of stress-induced self-proteins. The human MHC class I-like molecules MICA and MICB are examples of such inducible self-antigens. Stressed intestinal epithelial cells and some epithelial tumors induce their expression, and MIC molecules can be recognized by  $\text{V}\delta 1^+$   $\gamma\delta$  TCRs that are most prevalent among IEL (25, 26). The NK receptor NKG2D, which also binds to MICA and MICB (27), likewise is expressed by those  $\gamma\delta\text{TCR}$  IEL (28). It is thus possible that the rearranged receptors and the invariant receptors recognize the same molecules,

providing an opportunity for cross-talk between these receptor systems. Mice do not have structural homologs of MICA and MICB, although they have NKG2D receptors and class I-like molecules, including REA-1 and H-60, that can bind to them (29, 30).

The second major category of IEL are the  $\alpha\beta\text{TCR}^+$  cells that express the typical TCR coreceptors, CD4 or CD8 $\alpha\beta$ , although the ratio of CD8 $\alpha\beta$  to CD4 is much higher on these cells compared with the spleen cells. Conventional  $\alpha\beta\text{TCR}^+$  cells will migrate from the GALT and peripheral lymphoid tissues to the intestinal epithelium upon antigen stimulation in the periphery (31); these are essentially tissue-seeking effector or memory cells. Specialized M cells in the follicle-associated epithelium continuously sample the gut lumen and transport antigens without processing to the subepithelium and GALT (32). Local dendritic cells (DCs) then process and present these antigens and further distribute them via the thoracic duct (TD) into the bloodstream. Peripheral naïve  $\text{TCR}\alpha\beta^+$  T cells reactive to the specific antigens will then undergo activation and proliferation locally and migrate as antigen-experienced lymphocytes via the TD to the gut, from where they can seed the whole length of the small intestinal epithelium (33). Repeated identical antigenic stimulation progressively narrows the TCR repertoire of these conventional IEL, which consequently have an oligoclonal TCR repertoire (33). These cells, which have a memory function, have been shown to be protective against oral infections with several pathogens (34–38).

Gut-seeking, antigen-experienced T cells have some differences from those in the spleen, including an initial dependence upon CD40-CD40L interactions (36), effector function that can readily be measured *in vitro* (38), and longer survival than the memory cells in the spleen (38). It has recently become apparent, however, that antigen-experienced T cells migrate to a variety of tertiary sites, including the lung, liver, and intestine (39). This has led to the concept that there are two types of memory T cells that can be distinguished by expression of CCR7 and other molecules (40). Central memory cells are localized to lymphoid organs, whereas effector memory cells are found in the effector organs.

Although it is not fully understood what the specific signals are that determine the localization of these conventional T cells activated in the periphery to the intestinal epithelium, integrins containing a  $\beta 7$  subunit are important, and  $\beta 7$ -deficient mice have greatly reduced numbers of T cells throughout the mucosa (41–43). Chemokines are also likely to play an important role. The chemokine CCL25 (TECK) is produced by epithelial cells in the small intestine, but not by those in the large intestine (44, 45), and its receptor CCR9 likewise is found predominantly on small intestine IEL and LPL (46). Recent data suggest that this chemokine may be involved in the selective homing of conventional T cells to the small intestine (47), although the *in vivo* blocking of CCL25 with antibodies (48), or the analysis of CCR9 $^{-/-}$  mice (49), indicated only modest decreases in IEL. It is therefore likely that there is redundancy in this chemokine/chemokine-receptor system. Nevertheless, the exposure to antigens that are present in mucosal lymphoid structures is important for the migration of conventional peripheral

T cells to mucosal tissues. CD4<sup>+</sup> T cells activated in peripheral lymph nodes have low expression of the  $\alpha_4\beta_7$  integrin important for homing to the intestine, whereas those activated in intestinal lymphoid structures selectively express high levels of  $\alpha_4\beta_7$  and migrate in response to CCL25 (50). Similarly, for CD8<sup>+</sup> T cells, it has been shown that activation with DCs from PP induces  $\alpha_4\beta_7$  and CCL25 responsiveness, whereas activation of splenic or lymph node DC does not (51), suggesting an imprinting of a mucosal-seeking phenotype by PP DCs.

The third category of IEL includes TCR  $\alpha\beta^+$  DN cells, and the more numerous CD8 $\alpha\alpha$  single positive (SP) IEL. Like the  $\gamma\delta$ TCR IEL, the expression of a number of surface proteins characteristic of conventional T cells are reduced or lacking on these cells, including Thy-1 (52), CD28 (53), and CD2 (54). On the other hand, they exclusively express NK receptors and they have been dubbed "gut NKT cells" (55). Additionally, and typical of NKT cells, cells that express a canonical human V $\alpha$ 7.2-J $\alpha$ 33 or mouse V $\alpha$ 19-J $\alpha$ 33 TCR have been identified among the mucosal lymphocytes, particularly among the LPL. These mucosal-associated invariant T cells are selected and/or restricted by a nonclassical class I molecule, MR1 (56). The invariant TCR  $\alpha$  chain and reactivity to a nonpolymorphic class I molecule are features similar to the CD1d-dependent NKT cells, suggesting that this MR1-reactive population might constitute an intestinal-specific NKT cell subset.

CD8 $\alpha\alpha$  IEL preferentially use the invariant signaling component Fc $\epsilon$ RI $\gamma$  chain (57, 58), which also is a subunit of the multimeric Fc $\gamma$ RIII A (CD16) complex displayed by NK cells. The Fc $\epsilon$ RI $\gamma$  chain is expressed as part of their CD3 complex, either as a homodimer or as a heterodimer together with CD3 $\zeta$ . The presence of a different TCR signaling module, as well as the absence of coreceptor molecules and accessory/costimulatory molecules such as CD2 or CD28, indicate that the TCR-mediated selection and activation events for CD8 $\alpha\alpha$  TCR $\alpha\beta^+$  IEL are likely to be different from conventional T cells. Although CD8 $\alpha\alpha$  SP TCR $\alpha\beta^+$  IEL express an oligoclonal TCR repertoire, this repertoire does not overlap with the one expressed by the CD8 $\alpha\beta$  TCR $\alpha\beta^+$  IEL (59, 60). Furthermore, although potentially self-reactive TCRs are depleted from the peripheral repertoire during negative selection, these TCRs accumulate among the CD8 $\alpha\alpha$ SP TCR $\alpha\beta^+$  IEL (1). This implies that the CD8 $\alpha\alpha$  SP TCR $\alpha\beta^+$  IEL have not been submitted to conventional negative selection, and that the gut-specific CD8 $\alpha\alpha^+$  IEL must have originated under distinct selection pressure with a distinct ontogeny.

## T CELL LYMPHOPOIESIS AND THE INTESTINE

Even before TCRs had been discovered, based on the presence of IEL in neonatally thymectomized mice, Fichtelius postulated that the small intestine is a "first-level," or primary, lymphoid organ (61). Direct evidence for in situ development of T cells within the gut epithelium was provided by Ferguson & Parrott, who engrafted fetal small intestine tissue under the kidney capsule of athymic mice (62). They showed

that host lymphocytes developed to some extent in the intestinal grafts. With improved technology and new tools, other approaches were taken to provide further evidence to demonstrate and characterize T cell lymphopoiesis in the absence of a functional thymus. Although the results from a number of experiments provided evidence for an extrathymic origin for some IEL (63–65), in no case could it be demonstrated that functional  $\text{TCR}\alpha\beta^+$  T cells had been subjected to a positive selection process. The results showed that the extrathymic pathway of development was inefficient and mostly skewed to the  $\text{TCR}\gamma\delta^+$  population. Consistent with this thymus dependence, thymic transplant gives rise to IEL in chickens (66, 67), neonatal thymectomy causes a drastic decrease in IEL in mice (68), and early thymectomy results in decreased IEL in *Xenopus* (69). Congenitally athymic (*nu/nu*) nude mice, which have a mutation in the winged-helix transcription factor, *Whn*, that is required for differentiation of thymic epithelial cells (70), have greatly reduced  $\text{TCR}\alpha\beta^+$  IEL, although there is a residual  $\gamma\delta\text{TCR}^+$  IEL population (71). The level of  $\text{TCR}\alpha\beta$  and CD3 expression on the few  $\text{TCR}\alpha\beta^+$  T cells in nude mice is low, and the percentage of splenocytes expressing CD4 or CD8 $\alpha\beta$  is greatly variable among individual nude mice (72). Furthermore, the repertoire of  $\alpha\beta$  TCRs in nude mice is pauciclonal (73, 74) and often shows extreme bias toward certain V $\beta$ s (73). In minor lymphocyte-stimulating (Mls)-1a or I-E $^+$  strains, T cells expressing a V $\beta$ 6 or V $\beta$ 11 TCR are present in the periphery, whereas thymocytes expressing these TCRs in euthymic mice of the same strains are deleted during the process of negative selection in the thymus (73, 75). This provides further evidence that the  $\text{TCR}\alpha\beta^+$  T cells that develop in the absence of a thymus do not undergo the selection events characteristic of thymic development.

Evidence for lymphocyte development in the adult human intestine is scarce (76). mRNA for RAG-1 and RAG-2 could be isolated from IEL but not from LPL (77) while terminal deoxynucleotidyl transferase was undetectable (78). Nevertheless, adult IEL TCR $\beta$  mRNAs contain numerous N region additions (79, 80), indicating that any role in situ for RAG in building an IEL-specific  $\text{TCR}\alpha\beta$  repertoire must be very small. Human fetal intestine contains T cells in the LP and epithelium from 12 to 14 weeks into gestation (81), but the fetal thymus is already exporting T cells at this stage of gestation. Immunohistochemical staining of tissue sections of fetal intestine indicates that approximately 50% of the CD3 $^+$  IEL express CD8 $\alpha\alpha^+$ , whereas the rest are predominantly DN cells. The LP also contains numerous CD3 $^+$  cells, but CD4 $^+$  T cells outnumber the CD8 $^+$  LPL regardless of gestation age, and there are large numbers of DN cells as well (76). The majority of fetal IEL already express an activated phenotype, and unlike adult IEL, they are actively proliferating. It is unlikely that IEL in the fetal intestine have migrated there as effectors in response to nonself antigens. Together with the fact that fetal IEL, unlike IEL of the adult intestine, display a very diverse TCR V $\beta$  repertoire (76), it seems logical to suggest that fetal IEL represent a naïve repertoire. The antigen-experienced phenotype of the fetal IEL together with the fact that most fetal CD8 $^+$  IEL express CD8 $\alpha\alpha$  (82) is consistent with the finding in mice that CD8 $\alpha\alpha$   $\text{TCR}\alpha\beta^+$  IEL have acquired this phenotype during self-agonist-based

selection in the thymus (discussed below). In summary, the evidence from several species is consistent with a thymic origin of all or most IEL, and it is also consistent with the hypothesis that efficient positive selection of T cells occurs only in the thymus.

## LYMPHOCYTE PRECURSORS AND THE INTESTINE

Although abundant data support a thymic origin for IEL, the intestine retains some vestigial function in lymphopoiesis. In an immunohistochemical search for anatomical sites of lymphopoiesis within the gut, small clusters of lymphoid precursors called cryptopatches (CP) were discovered within the LP compartment adjacent to the epithelial crypts in the small intestine in mice (83). Similar structures in other species have been ambiguously described (84). In nude mice, none of these clusters contain CD3<sup>+</sup> TCR<sup>+</sup> cells, and the c-kit<sup>+</sup> cells located there were negative for expression of CD3, B220, Mac-1, Gr-1, and TER 119 (lin<sup>-</sup>) (83). Transplantation of CP subsets isolated from athymic nude mice to irradiated severe combined immune deficient (SCID) recipient mice indicated a lymphopoietic capacity for the c-kit<sup>+</sup> CP subset (83). Most c-kit<sup>+</sup> CP cells also expressed the IL-7R (85), and in both IL7R $\alpha$ <sup>-/-</sup> mice and common cytokine receptor  $\gamma$ -chain-deficient mice (CR $\gamma$ <sup>-/-</sup>), formation of CP was defective (86).  $\gamma\delta$ TCR<sup>+</sup> IEL, which are dependent on IL-7-R signaling for their development, were also absent in these CP-deficient strains. Consistent with a CP-independent origin, however,  $\alpha\beta$  TCR<sup>+</sup> IEL were still present in the CP-deficient strains but absent in athymic nude CR $\gamma$ <sup>-/-</sup> mice (86). TCR<sup>-</sup> IEL isolated from these athymic CR $\gamma$ <sup>-/-</sup> mice have characteristics resembling B220<sup>+</sup> fetal liver progenitors (87). Furthermore, although the adoptive transfer of CP cells to SCID recipient mice suggested the presence of T cell precursors, when transferred to whole-lobe fetal thymic organ cultures, CP cells showed no such capacity (83).

T cell precursor activity can be revealed among IEL as well. Co-culture of CD3<sup>-</sup>CD8<sup>-</sup>Ig<sup>-</sup>CD45<sup>+</sup> IEL with newborn thymic stromal cells, which had been depleted of CD45<sup>+</sup> thymocytes, led to the generation of DP (CD4<sup>+</sup>, CD8 $\beta$ <sup>+</sup>) and SP CD4<sup>+</sup> or CD8 $\alpha\beta$ <sup>+</sup> cells (88). DP cells that were generated from these co-cultures of intestinal progenitors and thymic stromal cells expressed  $\alpha\beta$ TCRs, whereas DN cells expressed either  $\alpha\beta$ TCRs or  $\gamma\delta$ TCRs. Furthermore, a significant amount of the T cells expressed NK markers (88). A subset of the TCR<sup>-</sup> putative progenitors expresses low levels of CD45 (CD45<sup>low</sup>). Cells with this phenotype are abundant among the epithelial cells of nude mice. The CD45<sup>low</sup> subset also expressed markers typical of immature thymic precursors, such as CD44<sup>high</sup>, CD16, CD24 (heat-stable antigen), c-kit, CD127 (IL-7R $\alpha$ ), and CD122 (IL-2/15R $\beta$ ) but lacked CD25 (IL2R $\alpha$ ). The presence of the IL-7R $\alpha$  and IL-2/15-R $\beta$  is consistent with a role for IL-7 and IL-15 in the ontogeny of IEL (89, 90). Cocultures of these CD45<sup>low</sup> CD3<sup>-</sup>CD8<sup>-</sup> cells isolated from nude mice with thymic stromal cells from normal mice also led readily to the development of DP, SP, and DN T cells similar to the developmental stages of thymocytes (88).



Studies on lymphopoiesis within the intestine are hampered by the dominant presence of mature T cells in the adult gut, as opposed to mostly immature precursors in the thymus. Using multiple lineage markers, progenitor cells isolated from CPs and IEL were further characterized and divided into six types (91, 92). Bone marrow from nude mice was transferred to thymectomized RAG-2<sup>-/-</sup> and CR $\gamma$ <sup>-/-</sup> double-deficient recipient mice to establish the sequence of appearance of these precursors during lymphoid development (92). CP lin<sup>-</sup> cells showed very little evidence for T cell commitment, but they expressed GATA-2, a transcription factor found in pluripotent hematopoietic progenitor cells (93). It is therefore possible that CP cells represent predominantly precursors of other hematopoietic cells.

Using several gene mutations that affect T cell differentiation, including pre-T $\alpha$ <sup>-/-</sup>, TCR $\alpha$ <sup>-/-</sup>, and CD3 $\epsilon$ <sup>-/-</sup>, it was shown that none of these mutations affected the CP cells, further demonstrating that these cells are not committed to the T cell lineage (92). IEL progenitors were more committed to the T cell pathway, and the TCR $\alpha$ <sup>-/-</sup> mutation led to accumulation of DP IEL. On the other hand, TCR $\delta$ <sup>-/-</sup> did not block the cells at the DP stage, and it is thus more likely that TCR $\gamma\delta$  IEL developed before the DP stage. Such a scenario is typical of the developmental pathway for  $\gamma\delta$  T cells in the thymus (See Figure 1). Pre-T $\alpha$  mRNA, uniquely expressed by the precursors committed to the TCR $\alpha\beta$  lineage in the thymus, was extremely low among the Lin<sup>-</sup> gut cells, further supporting the notion that T cell precursors in the gut are predominantly committed to the TCR $\gamma\delta$  T cell lineage (92).

To determine if the extrathymic pathway of T cell development is active in euthymic mice, transgenic mice that carry a reporter gene (green fluorescent protein, or GFP) driven by the RAG-2 promoter were analyzed (15). Cells expressing high levels of GFP in these mice indicate recent transcription of RAG-2, reflecting TCR and Ig gene rearrangements (94). Using this reporter system, it was shown that athymic mice show receptor rearrangements predominantly in the MLN and PP. CP cells did not express RAG-2, indicating that rearrangements of TCR genes did not occur in these cells. Mature cells migrated from the MLN to the TD, and eventually to the gut mucosa. Interestingly, this pathway was totally suppressed in euthymic mice, except under conditions of severe lymphopenia. Nevertheless, in euthymic TCR $\beta$ <sup>-/-</sup> mice, which only have mature  $\gamma\delta$  T cells, GFP<sup>+</sup> cells were present in MLN and PP (15). This indicated that the lack of extrathymic T cell development in euthymic mice is not due merely to the presence of a thymus, but rather the thymus-derived TCR $\alpha\beta$  T cells themselves actively down-modulate lymphopoiesis in the gut. The suppression is less pronounced in older mice where newly generated TCR $\alpha\beta$  T cells are much reduced due to the involution of the thymus (15).

## CD8 $\alpha\alpha$ TCR $\alpha\beta$ IEL ARE SELF-REACTIVE

Conventional selection of TCR $\alpha\beta$  T cells in the thymus is central in establishing self-MHC restriction (95–98) and self-tolerance (99, 100). The naïve TCR $\alpha\beta$  repertoire in the periphery is thus imprinted by the selective interactions of TCRs

with self-antigens/MHC during this selection process in the thymus (101). The affinity/avidity-based model for selection proposes that TCR signal strength is central to the outcome of thymic selection, with intermediate TCR signal strength resulting in positive selection (Figure 2) and further maturation of the  $CD4^+CD8\alpha\beta^+$  double positive (DP) immature thymocytes to the SP mature stage (102). Strong TCR-mediated selection signals delete those TCRs from the repertoire of conventional T cells (see Figure 2).

The  $CD8\alpha\alpha$  TCR $\alpha\beta$  IEL subset, however, is governed by different avidity requirements than those outlined above for conventional T cells. In MHC class II, I-E<sup>+</sup> mice bearing the Mls-1<sup>a</sup> allele of the Mls locus, T cells expressing TCRs (V $\beta$ 6, V $\beta$ 8.1, or V $\beta$ 11) reactive to the endogenous retroviral superantigen of the Mtv-7 provirus are deleted from the conventional repertoire while these TCR are present, and in some cases at frequencies higher than the nondeleting strain, in the  $CD8\alpha\alpha$  TCR $\alpha\beta$  IEL subset (1). The notion that  $CD8\alpha\alpha$  TCR $\alpha\beta$  IEL do not follow the rules of conventional selection was also apparent in TCR $\alpha\beta$  transgenic mice that also express the cognate antigen recognized by the TCR transgene. H-Y TCR transgenic mice, with a TCR that recognizes a Y chromosome-encoded *Smcy* peptide presented by D<sup>b</sup>, provide a paradigm for these types of studies. When the TCR transgene was crossed on a RAG<sup>-/-</sup> background (103), it was shown that  $CD8\alpha\beta$  SP thymocytes matured in the thymus and populated the peripheral tissues of H-Y antigen-negative female mice. There were few IEL in these female mice, however, which is consistent with the belief that conventional T cells migrate to the intestine only upon antigen stimulation. This stimulation might be provided to some extent by endogenous peptides of the kind that are responsible for the homeostatic maintenance of naïve T cells in the periphery. The H-Y TCR<sup>+</sup> T cells are unusual in that they do not undergo this type of homeostatic maintenance in female mice (104, 105), perhaps reflecting the low affinity of this TCR for endogenous peptides in female mice. In sharp contrast, male H-Y TCR transgenic mice developed large numbers of TCR<sup>+</sup> IEL, and they predominantly expressed  $CD8\alpha\alpha$ , despite deletion of conventional  $CD8\alpha\beta$  H-Y TCR<sup>+</sup> T cells (103, 106, 107). The active generation of the  $CD8\alpha\alpha$  TCR $\alpha\beta$  IEL under deleting conditions for the conventional TCR $\alpha\beta$  T cells clearly demonstrated that a different outcome pertained to the  $CD8\alpha\alpha$  TCR $\alpha\beta$  IEL with regard to self-recognition during selection. Similar observations were made in a number of other TCR transgenic systems expressing endogenous cognate antigen, including the 2C TCR system recognizing L<sup>d</sup> as an alloantigen (108), and in double transgenic mice expressing a transgenic TCR and the cognate antigen as a transgene, including OT-I/RIPmOVA (107) and F5 TCR transgenic mice expressing influenza virus nucleoprotein (NP) peptides as transgenes (109).

The selection of these  $CD8\alpha\alpha^+$  IEL is most effective in the presence of high-affinity self-antigens. This was demonstrated using the clonotypic TCR F5. The NP68 peptide is recognized by the F5 TCR with high affinity and induces profound deletion of thymocytes in vitro, whereas the peptide NP34 is a peptide recognized with lower affinity. NP68-F5 TCR double transgenic mice generated large numbers

of CD8 $\alpha\alpha$  TCR $\alpha\beta$ <sup>+</sup> IEL (109). Likewise, CD8 $\alpha\alpha$  IEL expressing the 2C TCR were readily generated in mice expressing the L<sup>d</sup> high-affinity ligand, whereas CD8 $\alpha\beta$ <sup>+</sup> T cells developed when the low-affinity ligand presented by H-2K<sup>b</sup> was present. Similarly, H-Y TCR<sup>+</sup> CD8 $\alpha\alpha$  IEL were more efficiently generated in D<sup>b/b</sup> homozygous male mice than in their D<sup>b/d</sup> heterozygous counterparts (110).

The expression of CD8 $\alpha\alpha$  on these IEL does not necessarily imply MHC class I restriction of the TCR. A similar, high-affinity self-antigen-dependent development of CD8 $\alpha\alpha$  expressing IEL also was observed using several MHC class II-restricted TCR transgenic systems (107, 111). These observations indicate that CD8 $\alpha\alpha$  does not function as a traditional TCR coreceptor and that selection of the CD8 $\alpha\alpha$  TCR $\alpha\beta$  IEL might occur in a coreceptor-independent fashion. This was also supported by the observation that CD8 $\alpha\alpha$  H-Y TCR<sup>+</sup> IEL are actively generated in male mice that are deficient for CD8 $\beta$  expression, whereas the female H-Y TCR transgenic mice on the CD8 $\beta$ <sup>-/-</sup> background failed to generate mature H-Y TCR transgenic T cells (107).

Although CD8 $\alpha\alpha$  is most likely not directly involved as an MHC-interacting coreceptor in the selection of self-antigen-reactive TCRs, all CD8 $\alpha\alpha$  TCR $\alpha\beta$  IEL depend on some kind of MHC class I expression for their development and/or homeostasis. CD8 $\alpha\alpha$  TCR $\alpha\beta$  IEL, but not TCR $\gamma\delta$  IEL, are drastically reduced in  $\beta$ 2-microglobulin-deficient mice (112–114). By contrast, CD8 $\alpha\alpha$  TCR $\alpha\beta$  IEL are present in mice deficient for the transporter associated with Ag processing (TAP) gene, whereas conventional CD8 $\alpha\beta$ <sup>+</sup> MHC class I restricted T cells are nearly absent in these mice (112). The classical MHC class I molecules are highly TAP-dependent for their cell-surface expression, although some nonclassical class I molecules, including the thymic leukemia (TL) antigen and CD1d are not (115, 116). Similar to the TAP<sup>-/-</sup> mice, animals deficient for the classical class I molecules (K<sup>b-/-</sup>D<sup>b-/-</sup>) efficiently generated CD8 $\alpha\alpha$  TCR $\alpha\beta$  IEL, whereas the population of conventional CD8 $\alpha\beta$ -T cells was greatly reduced (117–119). The somewhat more efficient generation of CD8 $\alpha\alpha$  TCR $\alpha\beta$  IEL in K<sup>b-/-</sup>D<sup>b-/-</sup> mice, as compared with the TAP<sup>-/-</sup> mice, led to the suggestion that TAP-independent and TAP-dependent nonclassical class I molecules could promote the generation of the CD8 $\alpha\alpha$  TCR $\alpha\beta$  IEL. It has been reported that mice deficient in the nonclassical MHC class I molecule, Qa-2, which is for the most part TAP dependent (120), were much less efficient in generating the CD8 $\alpha\alpha$  IEL, suggesting Qa-2 reactivity for some CD8 $\alpha\alpha$  TCR $\alpha\beta$  IEL (121). By comparing BALB/c mice with the Qa-2<sup>null</sup> Bailey substrain (BALB/cByJ), we were unable to confirm these findings (L. Gapin, M. Kronenberg & H. Cheroutre, unpublished data).

The general rule that emerges from these studies is that CD8 $\alpha\alpha$  TCR $\alpha\beta$  IEL are self-reactive, but they are not selected for self-reactivity to a single self-antigen or a single antigen-presenting molecule. It is difficult, however, to estimate the fraction of these cells that is reactive to classical class I molecules, nonclassical class I molecules, or MHC class II molecules, considering that in the absence of one specificity, cells with another specificity may be expanded. The diverse specificities of these cells, together with their requirement for  $\beta$ 2m, indicates that

a function mediated by an MHC class I molecule other than peptide presentation to a TCR is required for the selection of CD8 $\alpha\alpha$  TCR $\alpha\beta$  IEL.

## CD8 $\alpha\alpha$ TCR $\alpha\beta$ IEL ARE THYMUS DERIVED

The efficient generation of the CD8 $\alpha\alpha$  TCR $\alpha\beta$  IEL, under conditions that caused negative selection in the thymus of CD8 $\alpha\beta$  T cells, was consistent with the view that the CD8 $\alpha\alpha$ -expressing cells were generated extrathymically. As discussed above, studies on the lymphopoietic capability of the small intestine indicated that IEL could develop locally in the absence of a functional thymus. Nonetheless, the local T cell development in the gut was biased toward the development of  $\gamma\delta$  T cells. This, together with the drastic reduction of CD8 $\alpha\alpha$  TCR $\alpha\beta$  IEL in nude mice and mice that were neonatally thymectomized, suggested that the thymus is required to efficiently generate all TCR $\alpha\beta$  T cells, including the IEL. This is also supported by the observation that thymus grafts from normal mice efficiently generated CD8 $\alpha\alpha$ -expressing TCR $\alpha\beta$  IEL in recipient nude mice (122).

Interestingly, it was shown that the different subsets were generated with different kinetics, and that whereas fetal thymus grafts or grafts from mice up to weaning age predominantly repopulated the intestine of the recipient mice with CD8 $\alpha\alpha$  TCR $\alpha\beta$  T cells, thymus grafts from older mice were more efficient in generating the conventional TCR $\alpha\beta$  T cell subsets (122). This was not necessarily due to the reduced ability of the adult thymus to generate CD8 $\alpha\alpha$  TCR $\alpha\beta$  IEL. Rather, it could result from the suppression of the selection and/or expansion of CD8 $\alpha\alpha$  TCR $\alpha\beta$  IEL by the increasing population of conventional TCR $\alpha\beta$  T cells. Support for this hypothesis is provided by the demonstration that adult thymus grafts isolated from male H-Y TCR donor mice on a RAG $^{-/-}$  background readily and efficiently repopulated the intestine of RAG $^{-/-}$  recipient mice with mature H-Y TCR transgenic CD8 $\alpha\alpha$  IEL (107). It is possible that in the absence of conventional positive selection, as in the case of the male H-Y TCR transgenic mice on a RAG $^{-/-}$  background, the generation of CD8 $\alpha\alpha$  TCR $\alpha\beta$  IEL will continue as long as a functional thymus is present.

## AGONIST SELECTION: A NOVEL PATHWAY OF T CELL DIFFERENTIATION

Although it is thought that negative selection is a major tool for the immune system to eliminate self-reactivity from the T cell repertoire (123), the death of autoreactive cells is not all encompassing. New evidence suggests that an agonist-based positive selection pathway permits the immune system to preserve and specifically select for self-reactive TCRs (Figure 1) (107). As outlined above, strong correlative evidence indicates that self-agonists are required to promote the development of CD8 $\alpha\alpha$ <sup>+</sup> IEL. Consistent with active selection for high-affinity-based TCRs, it was found that among the selected TCR<sup>+</sup> thymocytes in male H-Y TCR transgenic

mice on a RAG<sup>+</sup> background, with the potential to edit their TCR to escape negative selection, there was no evidence for editing. On the other hand, on a nonselecting background, numerous thymocytes underwent editing of the TCR $\alpha$  chain (124). Similar high-affinity selection conditions have been described for CD4<sup>+</sup> CD25<sup>+</sup> regulatory T cells (125) and NKT cells, including those that express an invariant V $\alpha$ 14 TCR and are CD1d reactive (126), and those with other specificities (127, 128). Some of these agonist-selected T cell subsets have regulatory functions, suggesting that agonist selection enables the immune system to make use of T cells that recognize self to control the responses of nonself reactive T cells.

The development and selection of self-reactive T cells is not an imperfection of the thymic selection mechanism, or a mere escape of these cells from conventional negative selection to create anergic or nonfunctional T cells. Evidence suggests that T lymphocytes selected by self-agonists require true ERK-dependent positive selection signals mediated by the TCR (107, 128). This was inferred by their requirement for the TCR $\alpha$  chain-connecting peptide domain or  $\alpha$ -CPM. This unique motif appeared with the first ancestral TCR $\alpha\beta$  receptor in jawed fish, and it has been conserved over 500 million years, underscoring its crucial function during TCR $\alpha\beta$  T cell development (129). The  $\alpha$ -CPM allows for efficient association of the CD3 subunits with the TCR, and imparts an ERK-mediated signaling function during thymocyte positive selection. TCR transgenic mice lacking the  $\alpha$ -CPM motif have impaired positive selection of conventional TCR $\alpha\beta$  T cells while negative selection still occurs (130). When TCR/Ag double transgenic mice that also had a mutated  $\alpha$ -CPM  $\alpha$  chain were analyzed for the presence of CD8 $\alpha\alpha$  TCR $\alpha\beta$  TCR transgenic IEL, we showed a much reduced population of these cells in the  $\alpha$ -CPM mutant mice, demonstrating that agonist selection is also a true ERK-dependent positive selection process (107). Together with the lack of dependence on CD8 $\beta$ , these data indicate that the agonist-driven selection of CD8 $\alpha\alpha$  TCR $\alpha\beta$  IEL is not an escape from conventional negative selection.

The concept of agonist-based selection is complementary to, and not mutually exclusive from, the deletion of thymocytes during conventional negative selection. The TCRs that are eliminated from the pool of conventional T cells might specifically accumulate in the repertoire of agonist-selected T cells. Such a prediction would indeed be consistent with the accumulation of the cognate transgenic TCRs among the CD8 $\alpha\alpha$  IEL of double TCR/Ag transgenic mice (107) as well as with the accumulation of self-reactive TCRs among the CD8 $\alpha\alpha$  IEL in normal mice (1). Furthermore, it remains to be determined if at very high affinities, agonist selection is no longer possible and negative selection is completely dominant (Figure 2). The data from multiple TCR transgenic systems suggests, however, that an affinity in which only negative selection is found may be difficult to reach.

As a result of the agonist-selection process, T lymphocytes differentiate into highly specialized cells with an antigen-experienced phenotype and regulatory functions. Mature agonist-selected T cells migrate preferentially to specific organs and tissues including the intestinal epithelium or liver, as opposed to the migration of conventional naïve T cells, which predominantly migrate to defined lymphoid

organs in the periphery (Figure 2). The factors that are responsible for the selective localization of CD4<sup>+</sup> Treg to lymph nodes and spleen, V $\alpha$ 14 invariant NKT cells to the liver, and CD8 $\alpha\alpha$  TCR $\alpha\beta$  IEL remain to be determined.

## AGONIST SELECTION AND SELF-ANTIGENS IN THE THYMUS

If self-reactive T cells were primordial, then the thymus might have evolved initially to accommodate the agonist selection of self-Ag-specific T cells reactive to polymorphic self-molecules. Recently, it has been shown that tissue-specific self-antigens are expressed by a subset of medullary thymic epithelial cells (131). The ectopic expression of these self-antigens is controlled by the transcriptional regulator, AIRE (132–134). The human autoimmune polyendocrinopathy syndrome type 1 is a multiorgan autoimmune disease resulting from loss-of-function mutations in the *AIRE* gene (132, 134), and AIRE-deficient mice also showed autoimmune disorders similar to the human disease. These observations suggested strongly that the AIRE-controlled ectopic expression of self-antigens in the thymus was central to establish self-tolerance (135). It was proposed that the absence of expression of these self-antigens during selection in the thymus resulted in the absence of effective negative selection and the appearance of uncontrolled autoreactive conventional TCR $\alpha\beta$  T cells (132, 133). This was illustrated in a model system, in which TCR transgenic T cells specific for a hen egg lysozyme (HEL) peptide were crossed to transgenic mice expressing HEL under the control of the rat insulin promoter. The normal induction of thymic tolerance in the TCR transgenic T cells to the insulin promoter-driven HEL was abolished when these mice were crossed onto an AIRE-deficient background (133).

We speculate that AIRE, and a transcription factor that controls AIRE expression (see below), might be required for the positive selection of self-specific CD8 $\alpha\alpha$  TCR $\alpha\beta$  IEL as well as the negative selection of conventional T cells. The observation that agonist-positive selection of specialized T cells, including the CD8 $\alpha\alpha$  TCR $\alpha\beta$  IEL, is based on recognition of self-Ag, and the fact that efficient agonist-driven positive selection occurs simultaneously with negative selection of conventional TCR $\alpha\beta$  T cells, suggests that the autoimmune pathology linked with defects in AIRE mutant mice and humans might reflect impaired agonist selection in addition to defective negative selection. The absence of self-Ag-based agonist selection might lead to reduced regulation of conventional T cells by these self-antigen-specific IEL and other regulatory T cells. The regulatory role of CD8 $\alpha\alpha$  TCR $\alpha\beta$  IEL is discussed below.

It has been proposed that mouse CD1d reactive NKT cells with an invariant V $\alpha$ 14 rearrangement, referred to here as V $\alpha$ 14i NKT cells, are selected by glycolipid self-agonists (126). Similar to the CD8 $\alpha\alpha$  TCR $\alpha\beta$ <sup>+</sup> IEL, these NKT cells have a natural memory phenotype, and interestingly, their IL-4 and IFN $\gamma$  gene loci are in the open configuration (136) and they become active as these cells mature in the

thymus. Although thymic epithelial cells are not required for the CD1d-mediated presentation of self-antigens during selection of V $\alpha$ 14i NKT cells, we and others have shown that a signaling pathway in a radio-resistant host cell, with the NF- $\kappa$ B-inducing kinase (NIK) activating the NF- $\kappa$ B family transcription factor RelB, is required for V $\alpha$ 14i NKT cell differentiation (137, 138). Furthermore, evidence suggests that this signaling pathway is required in thymic medullary epithelial cells. Interestingly, RelB<sup>-/-</sup> mice also display a significant decrease in the number of IEL, including the CD8 $\alpha\alpha$ -expressing TCR $\alpha\beta$  T cells (R. Shaikh, M. Kronenberg & H. Cheroutre, unpublished data) indicating that a RelB-dependent pathway is also crucial for the development, homing, or homeostasis of TCR $\alpha\beta$  CD8 $\alpha\alpha$  IEL.

The fact that these two agonist-selected T cell populations are defective in RelB<sup>-/-</sup> mice, while conventional T cells are numerous, has led us to speculate here that the agonist-dependent selection pathway is specifically impaired in the RelB<sup>-/-</sup> mice. Similar to the AIRE<sup>-/-</sup> mice, RelB<sup>-/-</sup> mice also failed to delete conventional T cells with a high affinity for self, and these mice also show a T cell-dependent autoimmune pathology (137, 139, 140). This suggests that the RelB-dependent contribution could be the same for both types of selection, the agonist-positive selection and the conventional negative selection. Interestingly, it has been shown that AIRE expression by thymic epithelial cells is RelB dependent, and RelB<sup>-/-</sup> mice show abnormal thymic medullary epithelial cells and complete absence of AIRE expression (134). It is thus possible that the autoimmunity displayed by RelB<sup>-/-</sup> mice could be attributed in part to the lack of RelB-dependent AIRE expression. Consequently, this would imply that defects in central tolerance, which lead to autoimmunity, are not solely the result of impaired conventional negative selection, but that concomitant defects in the process of agonist selection, and the resulting absence of self-specific CD8 $\alpha\alpha$  TCR $\alpha\beta$  IEL and other agonist-dependent regulatory T cells, might contribute to the severity of autoimmune conditions that are particularly prevalent in the intestine.

## AGONIST SELECTION AND DISTINCT FEATURES OF SIGNALING

It is not understood what drives thymocyte precursors to the agonist-selection pathway or at what point in thymopoiesis precursors divert to the different selection pathways. We can only speculate that the quality of the TCR signal received during the initial selection events differentiates precursors along distinct ontogenetic pathways. Nearly all CD8 $\alpha\alpha$  TCR $\alpha\beta$  IEL express the Fc $\epsilon$ RI $\gamma$  chain as part of the CD3 complex, whereas conventional TCR $\alpha\beta$  IELs express homodimers of CD3 $\zeta$  or hetero dimers of CD3 $\zeta$  and  $\eta$  (57), an alternatively spliced form of  $\zeta$ . The genes for CD3 $\zeta/\eta$  and Fc $\epsilon$ RI $\gamma$  are related and genetically linked. Analyses of gene-deficient mice have indicated that CD3 $\zeta/\eta$  is required for conventionally selected TCR $\alpha\beta$  IEL (CD8 $\beta^+$  or CD4), but less so for the agonist-selected CD8 $\alpha\alpha$  TCR $\alpha\beta$  IEL, consistent with their usage of the Fc $\epsilon$ RI $\gamma$  subunit (58, 141, 142).

The differing composition of the TCR signaling components between agonist-selected T cells and conventionally selected T cells suggests likely differences both in the downstream signaling cascades and in the developmental intermediates required to produce the different T cell types. Immature thymocytes at the DN stage initially express Fc $\epsilon$ RI $\gamma$  as part of their CD3 complex (143), and upon transitioning to the immature DP stage they replace the Fc $\epsilon$ RI $\gamma$  with CD3 $\zeta$ / $\eta$ . The fact that Fc $\epsilon$ RI $\gamma$  is retained in the TCR complex of the CD8 $\alpha\alpha$  TCR $\alpha\beta$  IEL, combined with the fact that these cells can be selected in a coreceptor-independent fashion, suggests the possibility that these agonist-selected T cells might be selected at the DN stage, deviating from the conventional pathway that requires a DP intermediate (see Figure 1). In many TCR/Ag double transgenic models that promote the development of the self-Ag-specific CD8 $\alpha\alpha$  TCR $\alpha\beta$  IEL, the coreceptor-expressing thymocytes (DP) that proceed along the conventional pathway of selection are greatly reduced in number (107), indicating that the development of the precursors of these agonist-selected T cells could occur in a CD4/CD8 $\alpha\beta$  coreceptor-independent way. This was directly demonstrated using the class II restricted DO-TCR transgenic system. DN transgenic T cells are selected under agonist conditions on an I-A<sup>b</sup> background. In H-2<sup>b</sup> CD4<sup>-/-</sup> mice expressing human CD2 under the control of the mouse CD4 promoter, selected mature DN DO-TCR transgenic T cells did not express the CD2 transgene, suggesting that the DO-TCR transgenic T cells had developed without passing the DP stage (144). The DN pathway was also suggested in a recent phenotypic analysis of thymocytes from H-Y TCR transgenic mice, where in the male transgenic mice the majority of the DN thymocytes have a mature phenotype, whereas the DN thymocytes from female transgenic mice are predominantly immature cells (111). The immediate precursors of DP thymocytes are the CD4<sup>-</sup> CD8 $\alpha\alpha$ <sup>+</sup> thymocytes (82). Although there are very few in normal adult mice, these precursors are numerous in the TCR/Ag double transgenic animals, and they constitute all the CD8<sup>+</sup> thymocytes in the fetal thymus (82). Even though controversy exists as to what extent these CD8 $\alpha\alpha$  SP thymocytes are mature immune-competent cells, it was shown that TCR cross-linking of thymocytes at the CD8 $\alpha\alpha$ <sup>+</sup> SP stage interfered with their transition to the DP stage (145). Consistent with their development along an alternative selection pathway, as opposed to being a dead end, the CD4<sup>-</sup> CD8 $\alpha\alpha$ <sup>+</sup> immature thymocytes express high levels of antiapoptotic factors of the Bcl-2 family, and they lack DNA fragmentation (82). The difficulty in detecting these precursors in the thymus of normal adult mice could be due to very low numbers of these intermediates, or a different kinetics of their selection process with rapid export of newly matured cells.

## AGONIST SELECTION AND CD8 $\alpha\alpha$ -TL INTERACTION

Several factors, including the differential use of signaling molecules and the absence of coreceptor contributions, might influence the TCR signals received by the agonist-selected thymocytes. At the immature DN stage, however, where Fc $\epsilon$ RI $\gamma$



is expressed and CD4 and CD8 $\alpha\beta$  coreceptors are not, other mechanisms must operate to allow for deviation of the immature precursors and maturation along the agonist selection pathway. CD8 $\alpha\alpha$  may provide a key for understanding this. The coordinate binding of CD4 or CD8 $\alpha\beta$  with class II- or class I-restricted TCRs, respectively, and their association with the p56<sup>lck</sup> tyrosine kinase, are two requirements for efficient positive selection. The coexpression of CD8 $\alpha\alpha$  with CD4 or CD8 $\alpha\beta$ , and the expression of CD8 $\alpha\alpha$  on MHC class II-restricted T cells, are inconsistent with a coreceptor function for CD8 $\alpha\alpha$  and suggest strongly that CD8 $\alpha\alpha$  expression mediates another function. A number of studies have demonstrated that CD8 $\alpha\alpha$  is inefficient as a TCR coreceptor, even though it is the  $\alpha$  chain of the CD8 molecule that associates with p56<sup>lck</sup> (146, 147), and it is the CD8 $\alpha$  chain ectodomain that interacts with the  $\alpha 3$  domain of the heavy chain of class I molecules (148, 149). For example, CD8 $\beta^{-/-}$  mice, which express CD4 and CD8 $\alpha\alpha$  at their DP immature thymocytes, do not efficiently select class I-restricted conventional T cells (150, 151). It has been shown that CD4 and CD8 $\alpha\beta$  associate with p56<sup>lck</sup> and help sequester it into lipid rafts together with the TCR, thus localizing the kinase in proximity of the TCR/CD3 complex (152). Furthermore, it was demonstrated using CD8 $\alpha$  transfection of CD4<sup>+</sup> class II-restricted T cells that expression of the coreceptor not involved in the activation complex sequesters active p56<sup>lck</sup> away from the activation complex, thus down-modulating the TCR-mediated signaling cascade (153). These observations are very consistent with the effects observed when endogenous CD8 $\alpha\alpha$  is expressed by T cells. Unlike CD8 $\alpha\beta$ , CD8 $\alpha\alpha$  is not internalized with the TCR from the surface of activated cells (154), and associated p56<sup>lck</sup> is inefficiently provided to the TCR complex in part because of the inability of CD8 $\alpha\alpha$  to effectively merge into lipid rafts. It is CD8 $\beta$  that promotes raft association of CD8 (155), although this can be partially overcome if CD8 $\alpha\alpha$  can bind to the antigen-presenting MHC class I molecule.

The TL antigen, a nonclassical MHC class I molecule that is abundantly expressed on the epithelial cells of the small intestine in mice (156), is a specific ligand for CD8 $\alpha\alpha$  (157). TL, however, is not a typical antigen-presenting molecule, and TL transgene expression in the absence of classical class I expression in TAP<sup>-/-</sup> mice does not promote the selection of CD8 T cells in the periphery (H. Cheroutre, unpublished data). Furthermore, the crystal structure of TL demonstrates a narrowed and completely occluded groove that cannot present peptides (158). These observations suggest a function for this MHC class I molecule that is different from antigen presentation. Direct evidence for this was provided by demonstrating that the TL interaction with its receptor CD8 $\alpha\alpha$  expressed by IEL mediates profound modulation of TCR activation signaling resulting in reduced proliferation and cytotoxicity of the activated T cells while promoting specific cytokine production (157). Furthermore, the TL-CD8 $\alpha\alpha$  interaction can mediate survival and further differentiation of the antigen-stimulated T cells (L. Madakamutil & H. Cheroutre, unpublished data).

The pronounced expression of this receptor-ligand pair on immature thymocytes besides the IEL suggests a possible function for CD8 $\alpha\alpha$ -TL not only in the maintenance and regulation of mature IEL, but also in the development and

selection of CD8 $\alpha\alpha$ -expressing thymocytes. We speculate that the interaction of CD8 $\alpha\alpha$  on immature thymocytes with its ligand TL could mediate the survival during agonist selection in the thymus of CD8 $\alpha\alpha$  TCR $\alpha\beta$  mucosal T cells.

## REGULATED IMMUNE RESPONSES IN THE INTESTINE

All TCR $\alpha\beta$  IEL subsets display functional and phenotypic traits of activated or memory T cells. Using adoptive transfer of naïve TCR $\alpha\beta$  transgenic lymph node T cells, it was shown that virus-encoded antigen, but not soluble antigens, could induce the generation of memory T cells in the LP and epithelium of the recipient mice (35–38). The mucosal memory T cells were retained locally and expressed specific adhesion molecules that differ from those expressed by memory T cells generated in the periphery (38). They showed functional characteristics of memory T cells, including rapid recall responses in *in vivo* and *ex vivo* lytic activity, which from a protective point of view are critical at interfaces such as the gut epithelium (38). These results therefore suggested that conventional T cells specific for nonself-antigens can differentiate into IEL following activation. These IEL represent a pool of antigen-specific memory T cells that can respond rapidly and effectively upon secondary exposure to their cognate antigen encountered in the intestine. IEL are surrounded by cells with antigen-presenting capability. Enterocytes have been shown to function as antigen-presenting cells (159–162), and DCs are widely spread throughout the LP and the intestinal epithelium. The epithelial DCs are specifically equipped to preserve the integrity of the epithelial barrier by expressing tight junction proteins, including occludin and claudin, which can establish rigid contact with the neighboring epithelial cells while protruding dendrites across the epithelium into the gut lumen (163). Furthermore, the antigen-dependent migration of conventionally selected T cells to the intestine also could be a protective measure to preserve an intact epithelial layer and prevent undesirable immune responses from nonself-specific naïve T cells against nonpathogenic agents abundantly present in the gut. In this context, it is interesting to note that the intestinal epithelial cells lack significant expression of costimulatory molecules, including the B7-1 and B7-2 molecules, which are required for efficient antigenic stimulation of the naïve T cells (164–166).

Unlike the conventionally selected TCR $\alpha\beta$  IEL, which have migrated from the periphery after encountering antigens in an organized lymphoid structure, there is no evidence that CD8 $\alpha\alpha$  TCR $\alpha\beta$  IEL or their immediate precursors have resided in lymph nodes. We propose that agonist selection in the thymus imprints these cells with a “natural” memory phenotype during development, including expression of cell surface molecules typical of mature IEL. For example, these agonist-selected thymocytes can express the mucosal integrin,  $\alpha_E\beta_7$ , which can mediate binding to E cadherin expressed by the epithelial cells (167–168) and might function to direct these agonist-selected T cells effectively to the intestine and/or tether them to the epithelial cells.

The intestine displays a pronounced tone of immune regulation. Nowhere else in the body is there more evidence that the immune system has evolved numerous mechanisms to prevent useless or self-destructive immune responses. The self-specific IEL subset has been conditioned during agonist selection in the thymus to remain immune quiescent in the antigen-rich environment of the gut. Using T cells from TCR transgenic mice that also expressed the cognate antigen during selection, it was shown that the CD8 $\alpha\alpha$  TCR transgenic IEL that express a TCR specific for a self-antigen were not destructive to the epithelium (107). By contrast, in a transfer model of colitis induction, which is induced by the injection of naïve (CD45RB<sup>high</sup>) CD4<sup>+</sup> T cells into immune-deficient mice, it was shown that these self-specific CD8 $\alpha\alpha$  TCR $\alpha\beta$  IEL exert a regulatory role not carried out by other IEL populations (169). These cells controlled the migration to the intestine and immune responses of the pathogenic CD4<sup>+</sup> CD45RB<sup>high</sup> T cells, and the regulation by the CD8 $\alpha\alpha$  TCR $\alpha\beta$ <sup>+</sup> donor IEL was contingent upon expression and recognition of their specific cognate self-antigen in the intestine of the recipient mice (169). The absence or defect of these self-specific regulatory IEL might be the underlining factor in the predisposition of the animal to uncontrolled immune responses and tissue destruction mediated by conventional T cells, as is observed in several models of experimental colitis. The TCR $\gamma\delta$  IEL were not shown to display suppressive regulatory activity in these transfer experiments, but these cells have been shown to secrete keratinocyte growth factor, which can stimulate repair of damaged epithelium and restore the integrity of the barrier (170).

The intestinal epithelial cells themselves play a key role in the immune regulation by expressing distinct surface molecules and restriction elements and secreting factors that allow them to uniquely modulate and differentiate regulatory subsets of mucosal T cells. Immune-modulatory cytokines, in particular TGF $\beta$  and IL-10, have been implicated as interrelating key regulatory factors (171–173), with TGF $\beta$  serving as the primary anti-inflammatory agent to down-modulate Th1 mucosal immune responses (174). IEL express receptors for growth factors that epithelial cells secrete, including stem cell factor, IL-7, and IL-15. The gut epithelial cells likely also express cell surface molecules that are specific ligands for regulatory receptors expressed by the IEL. This was demonstrated by the interaction of the TL antigen on epithelial cells with CD8 $\alpha\alpha$  expressed by the majority of the IEL (157). When CD8 $\alpha\alpha$  is engaged by TL, it is able to modulate TCR signals, thereby adapting the immune response of the IEL to an environment where maintaining the integrity of the epithelial barrier is a high priority. Unlike the clonal expansion and cytolytic activity of peripheral activated CD8<sup>+</sup> T cells, IEL activated in the presence of the TL-CD8 $\alpha\alpha$  interaction show reduced proliferation and cytotoxicity but enhanced cytokine production allowing for protection without destruction of the epithelial cell layer (157).

It is likely that other receptor and ligand pairs expressed by IEL and epithelial cells also play an important role in regulating mucosal immune responses. Antibodies to the human biliary glycoprotein (CD66a), a carcinoembryonic Ag-related molecule, could block the cytolytic activity of human IEL (175). This molecule

is expressed on epithelial cells as well as on IEL, and it can engage in homotypic interactions (176). Cross-linking of the  $\alpha_E\beta_7$  integrin, which may provide a mimic of the E cadherin interaction with this integrin, increased the TCR-mediated activation of human IEL (177). IEL, particularly the CD8 $\alpha\alpha$  TCR $\alpha\beta$  subset, have been shown to express activating and inhibitory NK receptors, including members of the Ly49 family, CD94/NKG2 and NKR-PI (55). Like CD66a in human IEL, some of these receptors have ITIM motifs and therefore could suppress overly exuberant IEL responses. Similar to the TL-CD8 $\alpha\alpha$  interaction, and perhaps to some of the other examples mentioned here, it is possible that these NK receptor-ligand interactions also deliver signals that can control and modify TCR activation signals in order to adapt IEL to provide protective functions without epithelial destruction. It is therefore likely that the TL-CD8 $\alpha\alpha$  interaction provides a paradigm for these types of lympho-epithelial immune-regulation in the intestine, and we are just at the beginning stages of understanding them.

## CONCLUSION

Mucosal T cells of higher vertebrates have uniquely evolved to adapt to the gut-specific environment that imposes extremely challenging conditions for defensive mechanisms. Through evolution, the mucosal immune system has diverted immensely to provide rapid and effective protection against innumerable pathogens concurrently with maintaining the integrity of the epithelial barrier, tolerating commensal micro-organisms, and allowing for efficient nutrient absorption. The evolution of the thymus as the organ responsible for  $\alpha\beta$ TCR<sup>+</sup> T cell education has relieved the intestine of much of its primordial function in lymphopoiesis. CD8 $\alpha\beta$  TCR $\alpha\beta$  IEL are a subtype of conventional memory T cells reactive to foreign antigens. By contrast, CD8 $\alpha\alpha$  TCR $\alpha\beta$  IEL are self-reactive natural memory T cells with diverse specificities. Self-specificity is selected for in the thymus—a developmental pathway that may endow these cells with the ability to regulate immune responses. T cell localization and function in the intestine is influenced by selection in the thymus, the site of antigen encounter in the periphery for conventional cells, integrin and chemokine receptor expression, and a dense network of invariant and adaptive receptor-ligand interactions that maintain a continuous communication between the mucosal T cells and their neighboring intestinal epithelial cells.

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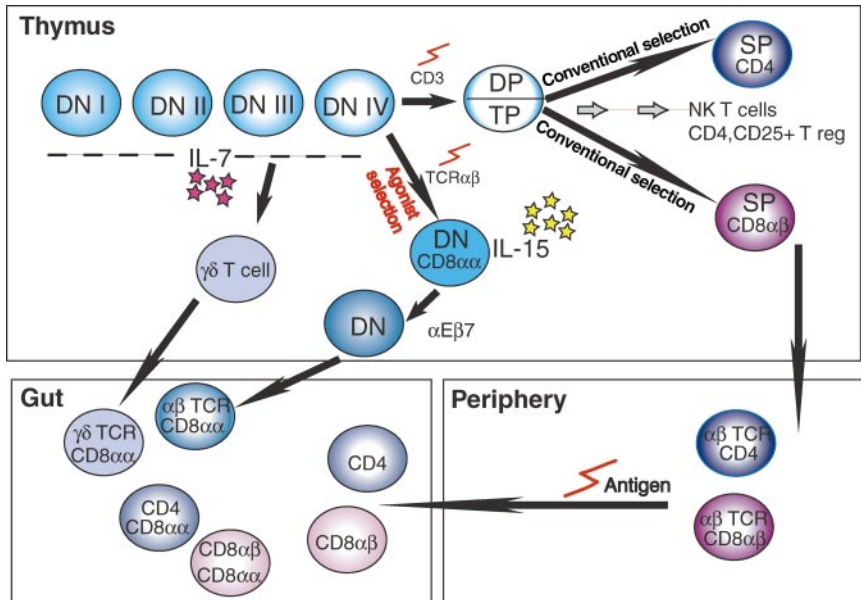
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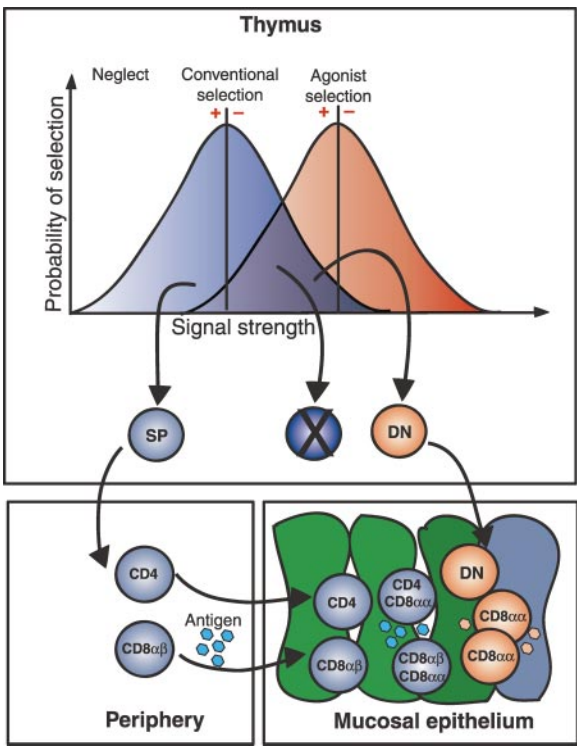
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**Figure 1** IEL are a diverse population of T cells. IEL consist of three major subsets: the TCRγδ T cells that are thymus derived and develop along the DN pathway; the conventional CD4 or CD8αβ TCRαβ IEL that have matured in the thymus along the conventional selection pathway and migrate as antigen-experienced T cells to the intestine; and the CD8αα TCRαβ IEL that have matured and differentiated in the thymus along the agonist-selection pathway and independent of coreceptor expression. The agonist-selection process directs their migration as self-antigen-experienced T cells to the intestine.



**Figure 2** Thymic selection is determined by TCR signal strength. Conventional positive selection occurs under conditions of intermediate TCR signal strength received upon recognition of self-MHC and self-antigen. Agonist selection occurs under strong signal strength conditions that otherwise delete thymocytes along the conventional negative selection pathway. Conventional selected mature T cells migrate to the intestine upon agonistic nonself activation signals received in the periphery.



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## ERRATA

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